Aufgaben und Lösungsvorschlag

Multiple Choice

MC1 [2 Punkte] Find the minimum on $(0, \infty)$ of the function

$$4x^2 + \frac{1}{x} - 1.$$

(A) 3

(B) 1

(C) 3/2

(D) 2

Lösung:

D The function goes to infinity at 0 and ∞, and the derivative is $8x - \frac{1}{x^2}$ that vanishes at $x = \frac{1}{2}$. Then the minimum is equal to

$$1 + 2 - 1 = 2.$$

MC2 [2 Punkte] For which $\alpha \in \mathbb{R}$ the following series converge:

$$\sum_{n=1}^{\infty} \sqrt{\frac{n \log(n)}{n^3 + n^{2\alpha}}}.$$

(A) $\alpha > 3/2$

(B) $\alpha \in (1, 2)$

(C) $\alpha < 3$

(D) $\alpha \in (0, 2)$

Lösung:

A It converges only when $2\alpha > 3$, so $\alpha > 3/2$.

MC3 [2 Punkte] Compute the radius of convergence of the following power series:

$$\sum_{n=1}^{\infty} \frac{n6^n}{3^{2n+1} + n2^n} x^n.$$

(A) 1/3

(B) 2/3

(C) 3/2

(D) 3

Lösung:

C The dominant term in the denominator is 3^{2n+1}. So

$$\sqrt[n]{\frac{n6^n}{3^{2n+1}+n2^n}} \sim \sqrt[n]{\frac{n6^n}{3^{2n+1}}} \to \frac{6}{3^2} = \frac{2}{3},$$

so the radius of convergence is $\frac{3}{2}$.

MC4 [2 Punkte] Compute the limit

$$\lim_{x\to 0} \frac{x+\cos(x)\sin(3x)}{\sin^3(x)+3\sin(x)-x}.$$

(A) 3
(B) 2
(C) 0
(D) 1

Lösung:

B The numerator behaves like $x+3x+O(x^2)$, while the denominator behaves like $3x-x+O(x^2)$, so

$$\lim_{x\to 0} \frac{x+\cos(x)\sin(3x)}{\sin^3(x)+3\sin(x)-x} = \lim_{x\to 0} \frac{4x}{2x} = 2.$$

MC5 [2 Punkte] Compute the integral

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}}\,dx.$$

(A) 2π
(B) π
(C) 2
(D) 1

Lösung:

B Let $u=\sqrt{x}$, then $x=u^2$, and $dx=2u\,du$. Also, when $x=0$ then $u=0$, and when $x=1$ then $u=1$. Substitute these into the integral:

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}}\,dx = \int_0^1 \frac{1}{\sqrt{u^2(1-u^2)}}\cdot 2u\,du = 2\int_0^1 \frac{1}{\sqrt{1-u^2}}\,du.$$

Now, we have a standard trigonometric integral. Let $u=\sin(\theta)$, then $du=\cos(\theta)\,d\theta$. When $u=0$ then $\theta=0$, and when $u=1$ then $\theta=\frac{\pi}{2}$. Rewrite the integral with the new variable θ:

$$2\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-\sin^2(\theta)}}\cdot\cos(\theta)\,d\theta = 2\int_0^{\frac{\pi}{2}} \frac{1}{\cos(\theta)}\cdot\cos(\theta)\,d\theta = 2\cdot\frac{\pi}{2} = \pi.$$

So, the value of the integral is

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}}\,dx = \pi.$$

MC6 [2 Punkte] Consider the first order ODE

$$u' = -\frac{1}{x}u + x^2, \quad u(1) = \frac{5}{4}.$$

Find $u(2)$.

(A) 2

(B) 5/4

(C) 5/2

(D) 3/2

Lösung:

C We write

$$u' + \frac{1}{x}u = x^2 \quad \Longrightarrow \quad xu' + u = x^3 \quad \Longrightarrow \quad (xu)' = x^3,$$

thus $xu(x) = \frac{1}{4}x^4 + C$. Imposing the initial condition we get $C = 1$ and therefore $u(x) = \frac{1}{4}x^3 + \frac{1}{x}$. In particular

$$u(2) = 2 + \frac{1}{2} = \frac{5}{2}.$$

MC7 [2 Punkte] Consider the second order ODE

$$u'' + 4u = 1 + 4x, \quad u(0) = 0, \quad u'(0) = 1.$$

Find $u(\pi)$.

(A) $2\pi - 1/4$

(B) $\pi + 4$

(C) π

(D) $2\pi + 1$

Lösung:

C The solutions to the homogeneous problem are $\sin(2x)$ and $\cos(2x)$. Also, a particular solution is given by $x + \frac{1}{4}$. So the general solution is given by

$$u(x) = A\sin(2x) + B\cos(2x) + x + \frac{1}{4}.$$

Imposing the boundary conditions, we get $A = 0$ and $B = -\frac{1}{4}$, hence

$$u(x) = -\frac{1}{4}\cos(2x) + x + \frac{1}{4}.$$

In particular, $u(\pi) = \pi$.

MC8 [2 Punkte] Let $(x_n)_{n=0}^{\infty}$ be a sequence. Which of the following statements is wrong?

(A) if $(x_n)_{n=0}^{\infty}$ is bounded, then the limsup is an accumulation point
(B) if $(x_n)_{n=0}^{\infty}$ is bounded, then there exists a converging subsequence
(C) if $(x_n)_{n=0}^{\infty}$ has no accumulation points, then it is unbounded
(D) if $(x_n)_{n=0}^{\infty}$ is a Cauchy sequence, then it is monotone

Lösung:

D

MC9 [2 Punkte] Let I be a bounded interval, and $f : I \to \mathbb{R}$ be a continuous function. Which of the following statements may be false?

(A) $f(I)$ is an interval
(B) $f(I)$ is open
(C) if f is bounded, then f is integrable
(D) if f is injective, then f is monotone

Lösung:

B

MC10 [2 Punkte] Which of the following improper integrals converges?

(A) $\int_0^\infty \frac{\sin(x)}{x^2}\,dx$
(B) $\int_0^\infty \frac{1}{x^2}\,dx$
(C) $\int_0^\infty \frac{1-\cos(x)}{x^2}\,dx$
(D) $\int_0^\infty \frac{1}{1+x}\,dx$

Lösung:

C

MC11 [2 Punkte] Which of the following sequences does not have a finite limit?

(A) $x_n = \sqrt{2n+1} - \sqrt{n}$
(B) $x_n = \sqrt{n^2+n} - \sqrt{n^2+2}$
(C) $x_n = \sqrt[3]{n^3+n^2} - n$
(D) $x_n = 2\sqrt{n+1} - \sqrt{4n+1}$

Lösung:

A

MC12 [2 Punkte] Which among the following sets is closed?

(A) $\{0\} \cup (1, \infty)$

(B) $\left\{\frac{1}{n} : n \in \mathbb{N},\, n \geq 1\right\}$

(C) $[0, 1) \cup (1, 2]$

(D) $\{2\} \cup (2, 3]$

Lösung:

D

MC13 [2 Punkte] Let $f : \mathbb{R} \to \mathbb{R}$ be a function and let $A, B \subset \mathbb{R}$ be subsets. Which of the following statements is correct?

(A) $f(A) \cap f(B) = f(A \cap B)$

(B) $f^{-1}(f(A)) = A$

(C) $f^{-1}(A) \cap f^{-1}(B) = f^{-1}(A \cap B)$

(D) $f(f^{-1}(B)) = B$

Lösung:

C.

Kästchenaufgaben

K1 [2 Punkte] Compute the indefinite integral

$$\int \frac{2x+1}{x(x-1)}\,dx.$$

Lösung:

Factor the integrand as

$$\frac{2x+1}{x(x-1)} = \frac{3}{x-1} - \frac{1}{x}.$$

Then

$$\int \frac{3}{(1+x^2)(4+x^2)}\,dx = 3\log|x-1| - \log|x| + C.$$

K2 [2 Punkte] Compute the indefinite integral

$$\int \frac{\sin(x)}{\cos(x)+1}\,dx.$$

Lösung:

Let $u = \cos(x)$, then $du = -\sin(x)dx$ and we get

$$\int \frac{\sin(x)}{\cos(x)+1}\,dx = -\int \frac{1}{u+1}\,dx = -\log(1+u) + C = -\log(1+\cos(x)) + C.$$

K3 [2 Punkte] Compute the Taylor expansion in 0, up to order 4, of the function

$$\sin(x^2) - \sin^2(x).$$

Lösung:

Since $\sin(y) = y - \frac{y^3}{6} + O(y^5)$, we have

$$\sin(x^2) - \sin^2(x) = x^2 - \frac{x^6}{6} + O(x^5) - \left(x - \frac{x^3}{6} + O(x^5)\right)^2 = x^2 - x^2 + 2x\cdot\frac{x^3}{6} + O(x^5) = \frac{x^4}{3} + O(x^5).$$

Beweise aus der Vorlesung

B1 [?? Punkte] Let $D \subseteq \mathbb{R}$ be a subset, $f : D \to \mathbb{R}$ be a function, and $\bar{x} \in D$. Show that the function f is continuous at $\bar{x}$ if and only if, for every sequence $(x_n)_{n=0}^{\infty} \subset D$ converging to $\bar{x}$, the sequence $(f(x_n))_{n=0}^{\infty}$ converges $f(\bar{x})$.

Lösung:

Suppose f is continuous at $\bar{x}$ and that $(x_n)_{n=0}^{\infty}$ is a convergent sequence in D with $\lim_{n\to\infty} x_n = \bar{x}$. Let $\varepsilon > 0$. Due to the continuity of f at the point $\bar{x}$, there exists $\delta > 0$ with

$$|x - \bar{x}| < \delta \implies |f(x) - f(\bar{x})| < \varepsilon$$

for all $x \in D$. Due to the convergence of $(x_n)_{n=0}^{\infty}$ to $\bar{x}$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies |x_n - \bar{x}| < \delta.$$

Thus

$$n \geq N \implies |f(x_n) - f(\bar{x})| < \varepsilon,$$

which implies that the sequence $(f(x_n))_{n=0}^{\infty}$ converges to $f(x_0)$.

For the converse, we assume that f is not continuous at $\bar{x}$. Then there exists $\varepsilon > 0$ such that, for all $\delta > 0$, one can find $x \in D$ with

$$|x - \bar{x}| < \delta \qquad \text{and} \qquad |f(x) - f(\bar{x})| \geq \varepsilon.$$

Using this for $n \in \mathbb{N}$ and $\delta = 2^{-n} > 0$, for each $n \in \mathbb{N}$ we find $x_n \in D$ with

$$|x_n - \bar{x}| < 2^{-n} \qquad \text{and} \qquad |f(x_n) - f(\bar{x})| \geq \varepsilon.$$

From these inequalities we conclude that the sequence $(x_n)_{n=0}^{\infty}$ converges to $\bar{x}$, but $(f(x_n))_{n=0}^{\infty}$ does not converge to $f(\bar{x})$.

B2 [?? Punkte] Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ a differentiable function. Show f is convex if and only if f' is increasing.

Lösung:

Remember that f is convex if for all $x \in (a, b) \subset I$ the inequality

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(x)}{b - x} \tag{1}$$

holds.

Suppose first that f' is increasing. Fix $a, b \in I$ with $a < b$, and consider $x \in (a, b)$. According to the Mean Value Theorem, there exist $\xi \in (a, x)$ and $\zeta \in (x, b)$ such that

$$f'(\xi) = \frac{f(x) - f(a)}{x - a}, \qquad f'(\zeta) = \frac{f(b) - f(x)}{b - x}.$$

Thus, from the monotonicity of f' we deduce (**??**), so f is convex.

Vice versa, suppose that f is convex. Let $a < b$ be two points in I, and consider $h > 0$ small enough so that $a+h < b-h$. Applying (**??**) on the interval $(a, b-h)$ with $x = a+h$ it follows that

$$\frac{f(a+h)-f(a)}{h} \leq \frac{f(b-h)-f(a+h)}{(b-h)-(a+h)}.$$

Also, from (**??**) applied on the interval $(a+h, b)$ with $x = b-h$, we get

$$\frac{f(b-h)-f(a+h)}{(b-h)-(a+h)} \leq \frac{f(b)-f(b-h)}{h}.$$

Combining the two inequalities above, we deduce that for all $h > 0$ sufficiently small,

$$\frac{f(a+h)-f(a)}{h} \leq \frac{f(b)-f(b-h)}{h}. \tag{2}$$

Taking the limit as $h \to 0^+$, we obtain $f'(a) \leq f'(b)$. Since $a < b$ are arbitrary, this proves that f' is increasing.

Anwendungen der Theorie

A1 [?? Punkte] Let $f : [0,1] \to \mathbb{R}$ be a continuously differentiable function.

a) Assuming that there exists $x_0 \in [0,1]$ such that $f(x_0) = 0$, prove that

$$|f(x)| \leq \int_0^1 |f'(s)|\, ds \qquad \forall\, x \in [0,1].$$

b) Assuming that $\int_0^1 f(s)\, ds = 0$, prove that

$$|f(x)| \leq \int_0^1 |f'(s)|\, ds \qquad \forall\, x \in [0,1].$$

and deduce that

$$\int_0^1 |f(s)|\, ds \leq \int_0^1 |f'(s)|\, ds.$$

c) Let $m = \int_0^1 f(s)\, ds \in \mathbb{R}$. Prove that

$$\int_0^1 |f(s) - m|\, ds \leq \int_0^1 |f'(s)|\, ds.$$

Lösung:

a) Use the fundamental theorem of calculus:

$$|f(x)| = |f(x) - f(x_0)| \leq \int_{x_0}^x |f'(s)|\, ds \leq \int_0^1 |f'(s)|\, ds.$$

b) Note that f cannot be always stricly positive or always strictly negative, hence by the intermediate value theorem there exists $x_0 \in [0,1]$ such that $f(x_0) = 0$. So we can apply (1) to get the first bound. Integrating the first inequality with respect to $x \in [0,1]$, we get the second inequality.

c) Apply (2) to the function $f - m$.

A2 [?? Punkte] Let $f : [0,\infty) \to \mathbb{R}$ be a continuous function, with $f(0) = 0$. Assume that $L = \lim_{x\to\infty} f(x)$ exists.

a) Prove that f is bounded.

b) Assume that $L \geq 0$. Prove that f attains its minimum.

c) Assume that $L = 0$. Prove that f attains both its minimum and its maximum.

d) Show that, if $L > 0$, then f may not attain its maximum.

Lösung:

a) Since $L = \lim_{x\to\infty} f(x)$ exists, there exists $R > 0$ such that

$$|f(x) - L| \leq 1 \qquad \forall\, x \geq R_1.$$

Also, by continuity, $|f|$ is bounded by M on $[0, R_1]$. Thus

$$|f| \leq \max\{M, |L| + 1\} \qquad \text{on } \mathbb{R}.$$

b) Let $m \in \mathbb{R}$ be the infimum of f. Since $f(0) = 0$, $m \leq 0$. If $m = 0$ then f attains its minimum at 0. If $m < 0$, choosing $\varepsilon = |m|/2$ we know that

$$|f(x) - L| \leq |m|/2 \qquad \forall\, x \geq R_2,$$

so, since $L \geq 0$,

$$f(x) \geq -|m|/2 \qquad \forall\, x \geq R_2,$$

This implies that the infimum of f is attained on $[0, R_2]$, and by the continuity of f on $[0, R_2]$ a minimum exists.

c) Apply (2) to f and $-f$.

d) Consider the function $f(x) = 1 - e^{-x}$. In this case, the maximum is equal to 1 and it is not attained.